ActInf GuestStream #013.1 ~ Adam Safron

GuestStream | Adam Safron | 1:45:37

hi i'm blue knight i'm here with daniel
friedman today is november 8th 2021
for a guest stream with adam saffron
adam please take it away
um
hi blue hi daniel hi everyone
it's good to be talking with you today
and
um i guess we're here to talk about
psychedelics and
the 5ht2a and related systems and
active inference
and so
i'll describe some recent
work on this and
looking forward to talking about it with
you
so let me get my screen
sharing
on
okay
coming through
excellent
so
um
i'm using my uh uh
active inference workshop slides as a
base and i'll go over those with a
little bit slower than i did uh in the
10 minutes i had there
and
then we'll take it from there and talk
it out
so
this is

you know part of an ongoing project to
both try to understand the
mechanisms of psychedelics but also the
impacted neurotransmitter
neuromodulator systems and the roles
they would have in active inference and
their potential significances for
their use as machine learning parameters
potentially
and so
when it comes to psychedelics um much of
the attention has been on the
5ht2a system and that's
a lot of evidence indicates that it
mediates the effects um
barely any work has been done
in machine learning on this and there
has been work done in active inference
and i'm going to introduce some a
compatible but slightly different take
on
the potential significances of these
systems
so
within machine learning a good amount of
work and with an active inference has
been done on the potential roles of
dopamine as a parameter
encoding
the
confidence one might have with respect
to policy selection
diane and others have suggested that
serotonin can act with an impotency to
dopamine based on a variety of lines of
evidence from neurophysiology and so
some work has also been done
in introducing a complementary

uncertainty parameter in the form of
serotonin this opponency can be thought
of um almost yin yang like with uh
dopamine being more of the yang and
serotonin being more of the yin
dopamine in general uh indicating in the
neuroscience literature's
incentive salience uh
confidence and if excessive potentially
impulsivity uh serotonin
would be
an opponency or an or complementarity i
would indicate the satisfaction of
desires states of uncertainty and
potentially a factor helping to
allow for patience rather than
impulsivity with respect to policy
selection
so
very little work though has been done in
machine learning uh on 5ht2a and also in
terms of active inference in terms of
actually uh modeling the 5hd2a system as
an active inference parameter
work is probably going to begin soon in
this direction
largely informed by the rebus paradigm
which we'll talk about soon
but
go back to uh the fundamentals of these
systems uh
the work that's been done so far with
the serotonin system can be thought of
more as relating to the 1a system
this would be
in contrast to the higher threshold i
believe it's a 5 to 1 ratio of
sensitivity uh

for the receptor of the two-way system
so when serotonin uh both of these by so
5ht is serotonin and you have some
i think 14 maybe 17 classes of receptors
depending on how you count them
but you know so how are we going to talk
about all of them well maybe eventually
but
it seems that
we might be able to go a fair ways uh
focus in terms of understanding their
significance
focusing on a few classes and sort of
working our way forward
with the additional receptors maybe
having a common significance to the
other ones but just
because they're located at different
parts of the brain it's a common
organism significance but functionally
you need them to do different things if
located in different places
into more detail soon but the 1a system
is shown to in general have more
inhibitory effects on
the signaling of pyramidal neurons while
the 5 hd2a system tends to be more
excitatory
carhart harris and nut have suggested
that
these both in a face of challenge
would be responses to challenge but the
1a might be something more like an
active or a passive coping response
being more
still
patient stop and think as opposed to the
2a system which might be more of an

active
coping response in the face of challenge
something more imaginative and
potentially creative
so
let's
yeah let's keep going
so
going back to dopamine versus serotonin
though um serotonin in general not
focusing necessarily on 1a versus 2a
there seems to be
some evidence suggesting that
dopamine will push you or actually a
good amount that dopamine tends to push
the organism towards more externally
focused attention in terms of uh helping
to increase the gain on more
exteroceptive and salience networks for
externally focused attention
5ht in contrast seems to be up
regulating the default mode and more
internally focused attention so that's
another interesting
difference
but to come to the 2a
so these receptors as i mentioned before
they tend to have
high thresholds for activation by
serotonin
1a will be modulating mood as their
serotonin levels are going up and down
you might get this sort of gentle
adjustment but 2a it's going to take
more to really get those systems going
in fact
some of their activity might be
modulated by things other than serotonin

so there's evidence that things like
carbon dioxide um
maybe blood acidity in general is
changing their
activity and uh maybe the endogenous dmt
system which we could talk about later
uh so
psychedelics tend to come in about two
classes uh
one or classic psychedelics now one
would be based on the structure
structure of tryptamine the precursor of
serotonin and the other would be based
on the structure of phenylethylamine uh
precursor of cachetolamines uh both of
these seem to be active at the five hd2a
system
which
a variety of lines of evidence have been
shown to mediate the effects of
psychedelics to a very substantial
degree
so
in the
uh machine learning
type work i've been collaborating with
um jara
shikhspahi
to try to
reverse engineer these different
neuromodulators as machine learning
parameters and so we've been
using an architecture which has elements
of hans nettuber's
world models
and uh danajar is a dreamer
to
interpret uh 5ht signaling as

potentially being a pr
as the different classes receptors as
parameters for imaginative planning or
an active inference terms a
sophisticated entrance
uh so
with respect to world models you'll see
is this working yeah it's working
so and in han schmidt hoover's
architecture they'll use these auto
encoders to take these different frames
this is in a doom game they'll compress
them into a lower dimensional latent
space and then they will unpack these
and
through this
training of the autoencoder to learn how
to reconstitute
these frames
you get this
low dimension source of
low dimensional vectors in latent space
which can be used for training and with
much better computational efficiency and
some learning properties um when you
don't have to go all the way back to
pixel space where things can get messy
and you don't know what's causing what
if you work with this latent space
you can
come out substantially ahead in terms of
training and so you have this
variational auto encoder which is
learning how to compress scenes so this
is one aspect
or compress the observations you then
feed this latent space
compressed vectors into this

recurrent network which learns to
predict the state transitions
you know shove this into a much lower uh
dimensional controller which uh they
trained uh with uh evolutionary
strategies and then used this for policy
selection
generating new observation and the
standard sort of um reinforcement
learning uh paradigm
uh
the interesting thing about these is
they could also be run offline so they
you're able to take these and uh take
the outputs and feed it right back in
and you can do imaginative rollouts so
you can use this for planning or for
having the
agent train in its imagination
and so this would be an example i
believe the left would be
but scene and or the left would be uh
the the compressed latent space and then
the right
would be the agent actually
i think the left is what was actually
seen this is pixel space and the right
is the the imaginative latent space
so
it tends to be you're not really seeing
it here but it tends to
these
imagine trainings like like imagination
relative to perception or sensation
it's less fidelity but you can still use
it for the sake of planning and learning
but one issue is that
these agents can do a kind of um

self-adversarial attack when they're
training their imagination
so if the the temperature the
uncertainty isn't high enough they can
uh basically come up with these sort of
fantasies of uh easy environments where
they you know are just winning but then
this doesn't transfer to the real world
and so then by introducing a temperature
parameter
to increase the uncertainty of your
imaginative rollouts um this allows for
better training back to the real world
and so this
um i've been suggesting is potentially
one interpretation
of the 5-ht-1a system um
the
as increasing the temperature um putting
the organism basically instead of
dopamine would be you're confident go
forth and act uh serotonin acting on the
1a system would say we're not so
confident stop and think or imagine
uh don't act right away but let's let's
think this out or let's imaginatively
plan this out
look a little bit in our mind's eye
before we leap
the two way though
seems like
it's almost like a hybrid of the 1a and
dopamine where it's like okay
imagine more
but it's not necessarily doing so under
a state of uncertainty within these um
imaginative rollouts things are
appearing more vivid where the gap

between imagination and reality uh seems
to be uh narrowing
and the idea is of if you're thinking
back like passive versus active coping
so you're uncertain let's say the 1a
system is in the mix now you're okay i'm
not going to overtly act i'm going to
stop and think i'm going to reduce the
gain on the um expected free energy or
the expected value of pursuing a policy
right now
patience but if the 2a gets in the mix
it's going to tend to be more active
you're going to be imagining
but
you're going to be more likely to act on
these imaginings because you're uh
they're more compelling uh more vivid
so
now we're coming back to uh psychedelics
but i want to lay out this foundation
because the whole
way through the method is basically
a kind of marian neurophenomenology
trying to understand the mind at
computational functional algorithmic and
implementational levels of analysis
and then mapping this on to
rich descriptions of experience of
subjective experience so
mapping standard the cognitive science
multi-level stack to phenomenology
and
so the machine learning is part of
pulling off this mapping we could also
or active inference is part of that
so
uh to rehearse um the basics of rebus uh

other am i coming across all right or
just checking in okay
so the
relax beliefs under psychedelics this
i'd say might be the dominant paradigm
for understanding the significance of
the five hd 2a system
in creating the effects of psychedelics
under a predictive processing paradigm
so
according to rebus
you'll get
i don't think i have to rehearse
predictive coding here but very briefly
uh the idea is you have a hierarchy
where
of predictions where you are passing
your priors or your prediction errors
down the hierarchy
back towards the primary modalities
where the observations are coming in and
then you're passing prediction errors
upwards
and this coding scheme is efficient
especially if the events that you're
tracking or trying to model are clocking
slower to the internal states so
with something like telecommunications
if all you have to do is uh
transmit what's different between the
frames and uh i don't know 24 hertz used
to be what tv was back in the day but if
most of the pixels are largely the same
frame to frame and all you do is you
show what's different you get this
energy savings and when you're dealing
with an organ as hungry as a brain two
to three percent of your body mass 20 of

the budget
that the savings could be substantial uh
and
nature world of the quick and the dead
natural selection might have picked such
a thing
so
within creative coding
the
predictions are thought to be encoded by
these uh so if you think of cortex as um
jeff hawkins says it's roughly uh a
dinner napkin and thickness an extent
that's been crumpled up and shoved into
your head cortex has these different
layers um roughly six
and so
and this is distinct from like levels of
a hierarchy but the cortex the sheet if
you go like across the napkin
there's these layers about six
and depending on where you are they're
more or less distinct but the l5 or the
deep pyramidal neurons these are the
ones that will leave cortex do things
like form loops with thalamus the
striatum uh cerebellum
spine
and
that's what will is thought to encourage
your predictions these l5 deep pyramidal
neurons not deep in terms of a deep
police hierarchy but deep in terms of uh
relative to the coracle surface
more in there um and within the napkin
uh the prediction errors are thought to
be encoded by the superficial pyramidal
neurons on layers two three so l5 is

forming these big um synchronous
complexes with the thalamus
probably not the only way it does the
predictions but this will be um
aggregating information from multiple
different places giving you
and because you're able to get a a lot
of different causes brought together
it's going to be a good source of priors
it's going to be a good source of your
expectation and your modeling
and then you use this to try to suppress
the ascending stream of prediction
errors along this superficial layer 2 3.
you can go into layer later if that's
useful
but the idea is that
with under psychedelics
you take l5 pyramidal neurons these
excitatory neurons and you excite them
so much with 5ht 2a stimulation that
they fall out of sync with each other
and it's um
it's a kind of paradoxical effect when
they fire too readily they don't
coordinate anymore in establishing a
synchronous rhythm and because of this
they're actually
paradoxically relaxing the priors become
less precise the uh belief or free
energy landscape becomes flatter
and so now the idea is with your
expectations being less
more information from the periphery is
capable of penetrating deeper into the
cortical hierarchy registering more in
consciousness and
impacting

learning and updating to a potentially
greater degree and so this is a model of
the much of the phenomenology of
psychedelics as well as their mechanism
of therapeutic action for helping people
to um
as pollen would say like how to change
your mind helping you to change and
change your mind to update
and so
um
i think rebus is uh largely correct but
i've been introducing a potential
refinement of it um in something i call
albus which includes in addition to
rebus effects uh something i call sevis
effects or strengthening beliefs and so
uh the idea is that
you might not get exactly this
desynchronization effect along the whole
dose response curve
of 5ht2a agonism
or stimulation of the 582 t2a receptor
rather this might be the case of what
you would expect from very high levels
of agonism um that's really strong
excitation but let's say we increase the
gain like just a little bit or a
moderate amount i suggest it's not at
all obvious that this would cause the
energy landscape to flatten this would
not cause them necessarily to fall out
of sync with each other in fact they
might be able to sync up even more
powerfully if you just increase it a
little bit
or moderately we don't actually know
this might vary from you know organism

to organism uh but the idea is we might
need um different regimes different
accounts depending on where we are with
what level of stimulation before we
would expect to see the kind of
paradoxical desynchronization
or potentially more straightforward
enhanced synchronous ability
the
yeah
so
to bring it back to predictive coding so
here if you're going to try to
illustrate this uh the purple would be
your deep pyramidal neurons uh
supposedly encoding
or that are thought to encode your
beliefs and then up top you have
the white circles this would be your
superficial pyramidal neurons the upper
part of the layer and this would be
encoding your prediction areas that
you're passing upwards and so you're
seeing these i'm illustrating the
synchronous complexes by these red
swaths the predictions going downwards
are these red arrows and then um i'm
depicting basically the prediction areas
as these gray swabs some
and black arrows so within predictive
coding there's been significance
assigned to different
eeg and meg frequency bounds as
potentially corresponding more or less
to predictions or prediction errors more
specifically the slower rhythms like uh
alpha and beta where alpha is maybe
roughly 8 to 12 times a second betas

maybe 13 to 30 times a second that these
slower rhythms would be the ones
encoding your predictions
uh pass downwards
but that your prediction errors might be
more likely to be encoded by
gamma frequencies you know anywhere from
40 to 120 even even higher uh
oscillations per second
uh so within this part of the reason
that this might make some intuitive
sense
is if you're thinking of
the formation of a synchronous complex
of neural activity
uh that
uh faster would be smaller and slower
would be bigger or more inclusive more
integrative this might make some
intuitive sense if you're thinking of
okay so you're trying to get this
agreement this negotiation of a common
rhythm of a bunch of signaling units
if you're having more units spread over
a bigger distance you're trying to get
them to go come into agreement this
might necessarily have to be a slower
process
but if you're let's say like quantizing
prediction error generally like a little
packet of like a local agreement that
you're passing upwards
well that you can potentially clock a
lot quicker um so
that's the basic idea there so
if we were to look at rebus it would
look something like you see uh the the
right bottom uh the

swath of red is more restricted in scope
and the arrows are a little bit smaller
and so this is idea of you've relaxed
your beliefs this is um if you get this
paradoxical um
desynchronization
but uh
and so once you have this you're seeing
less of these prediction areas are being
inhibited by this descending stream of
predictions and they're able to make
their way in more
deeper into the hierarchy and this would
be showing you know one modality
obviously the brain is not just one
singular hierarchy but it's a hierarchy
multiple intersecting hierarchies
corresponding to each of your modalities
all brought together stitched together
into a hierarchy
up top
would be illustrating a sebis effect and
so the swath is bigger
and the
arrows are also uh darker and bigger and
that's and more prediction errors are
being inhibited
there's an additional factor uh which is
five
ht2a
receptors are also found on the
inhibitory interneurons of layers two
three
and so in theory uh and there's some
evidence i believe that suggests this uh
this would correspond to a default
reduction in the gain on the ascending
stream of prediction errors or you would

say an active inference speak um
less precision weighting of the
precision
of the prediction errors and so they
have less of a contribution of the
bayesian updating that's happening at
each level of this belief hierarchy and
so that's illustrated by um slightly uh
uh i don't know if this is
yeah the on their own the arrows are a
little bit smaller so
the idea here with the cebus effect
would be not only might you have
stronger beliefs
but these beliefs might be further
shielded from contradiction by
inconsistent sense data
i think this potentially provides a
fairly parsimonious account of some of
the conditions under which you get
certain perceptual alterations of a
hallucinatory variety so for instance
people will sometimes report
hallucinations
in sensory deprivation tanks
they'll sometimes
i think it's called a charles bonnet
syndrome where as you start to let's say
go blind or deaf you'll start to get
like auditory or hallucinations within
the modality as you lose input from the
external world
uh this also you could think of at a
higher level i don't know if this works
but you could think of potentially
delusions in a similar way a belief
which is overly strong and resistant to
updating from uh potentially uh

contradictory and potentially correct
sense data so in theory for talking
about sebis effects you might have a
single parsimonious account
for both hallucinations and delusions
that's incorporating a little bit more
of the neuroanatomy
that being said i'm not saying at all
that rebus is incorrect i actually think
it's
probably very correct
and uh maybe crucially important for uh
explaining some of the really powerful
changes that people observe with
psychedelics as a therapeutic
intervention
so
get into a little more details here and
it's okay uh y'all with me still there
okay cool
so
i mentioned a header here earlier so
i've taken these um predictive coding
hierarchies here
and i'm just illustrating them as these
triangles and so here would be let's say
uh you're uh
so
i think in this framework of albus i'm
suggesting uh
we need to even uh in addition to
including things like sebis effects i
think we're going to need to take on
things like
basically
consciousness
and
higher order consciousness to actually

really adequately explain what we mean
by relaxed or strengthened beliefs and
what actually is happening with these uh
different neuromodulators and maybe
attempts at creating analogs
so
uh here you know up top would be somatic
sensation
to the right would be your vision and
then your hearing and we're going to try
to bring all these things together and
so if you're thinking of some sort of
hierarchy like here you might have like
small beta complexes you can think of
each of these as like a small modeling
effort and a small belief over whatever
is in the scope of this synchronous
complex
where the idea being that when you
synchronize neurons
this allows them to be aligned in their
activity and time such that their
signals can
sum and be integrated rather than decay
and so uh this would allow you to have
potentially a joint belief over whatever
is in the scope of a synchronous complex
uh pascal freeze calls this
communication through coherence
and uh for a reference for this idea of
like an inverse hierarchy of size and
speed butsaki would be a good reference
for that i'd say
so here we are on the left the
beginnings of modeling of
increased hierarchical depth
with
small fast beta complexes the extraction

of things like line segments and maybe
basic contours and vision uh basic uh
patterns within sound within touch
but then let's say we nest these within
maybe maybe slower broader more
encompassing beta
we could have
more hierarchical structure for our
modeling effort
and so here you might start to get the
beginnings of things like objects as you
bring together these different features
the beginnings of things like objects
with respect to touch your your or
rather your own body in relation to
those um the beginnings of things like
meaningful sounds
and so then
let's say we bring all the modalities
together now we let them cross talking
about multi-modal representation
that you've organized according to a
coherent perspectival reference frame
well now you might get the beginning of
something like phenomenality or
experience and so here there's a blue
swath corresponding to alpha
frequencies here um
where
at alpha frequencies that would roughly
be it looks like the size that you would
need so 8 to 12 times a second of a
frequency of updating or
or sampling or estimation that this
would be big enough to get you the
entire sensorium all brought together
bringing together that the
different hierarchies into a hierarchy

alpha has some additional significance
in that it has particularly concentrated
sources of alpha power from midline
structures
and
i take this as being important
because if you look at something like
the posterior midline we look at let's
say
the posterior cingulate cortex which is
heavily implicated in the psychedelic
and also the meditation
research literatures as when alpha goes
down there seems to be some associated
with phenomena such as ego dissolution
uh
but
the idea that alpha might be important
for
coherent
egocentric perspective
in addition to bringing together your
sensorium for this mutual information
these midline structures in particular
the posterior cingulate um there's this
old idea of the papez circuit where it's
this uh
core circuit for emotional memory um
that would roughly go
manulary bodies around the brainstem it
goes up through the mammolothalamic
tract i think anterior lateral thalamic
nucleus
goes through the cingulum bundle
posterior cingulate into the hippocampus
out through the foreign inks back to
bodies don't worry about those details
the important thing about this

is uh
this was interpreted um by um
actually i said papas i think it's papes
but it was interpreted by papes when he
was etching out these elect sources of
epilepsy form discharge as this core
circuit for emotional memory and
part of the reason
that and so the importance of this
is
uh the mammary bodies are receiving
stretch receptor information from the
neck
and so this is going to be potentially
important for knowing where your head's
pointing especially if you're talking
about like
which will be important for us and maybe
even more so if we're talking about like
our inner rodent or
or whatever that you know where your
head's pointing is going to really help
for bringing coherence to
explaining your sensorium uh also at
this islamic relay you're also getting
inputs from the vestibular apparatus of
the inner ear and so this is giving you
like the yaw pitch and roll and so not
only do you have your entire sensor in
all the information at these posterior
midline structures before they funnel
into the hippocampus as a source of new
memory not only do you have the entire
sensorium brought together you're on top
of it all in network terms
but you have the information you would
need to organize it
by your egocentric perspective

and so
this yeah so here we go predictive
coding we covered
uh sebisfx redus effects
and now
i'm introducing like a basic model of uh
perceptual synthesis where we have
something of a cartesian theater here
enabled by
um uh
the ability of slow rhythms of a large
enough extent to bring together the
right kinds of information you would
need
to
uh
basically create a coherent
uh world model
so y'all with me stellar
okay
so
uh in this preprint
um
i describe how we can maybe think of
with respect to either sensation
or imagination a different combinations
of rebus and sebas effects happening
potentially corresponding to different
levels of stimulation
of these receptor classes of
the 582ar in particular
by the way um
i think 5h
5 ht uh 2c
that's gonna that's one to watch well it
might even be like almost more strictly
rebus in its nature but that's something
to talk about later um

so for sensation on the left here
you know this would be there's a person
out in nature and they're having an
interaction with a tree
and then on the right here would be a
similar thing but this is imagination uh
maybe you know with your eyes closed
maybe not but this is you know um here
you're imagining yourself
uh and
interacting with the tree now you know
imagination it doesn't need to be in a
third person but i'm just taking
advantage of the fact that we're talking
like if you are to think of yourself in
a third person if you're going to move
from this um
was it the
as james called it like from the eye to
the me
uh to this objectified selfhood that
part is going to require imagination you
can imagine yourself from the first
person but the third person point of
view um you're gonna require um
imagination
i guess that or a mirror or a good setup
of mirrors maybe you could do it that
way
so that actually might not that might be
non-trivial but on the left here
so the top we have let's say unaltered
if such a thing ever exists we're always
altered in some ways we're always on
some
our brain stems pharmacopoeia is always
dripping some combination of drugs to us
it's just uh sometimes it's a very

different regime and you know we say
it's altered but there is no like
singular baseline you know we who knows
how much we both vary as individuals and
across individuals
along these lines
but
so as we move down
let's say
we were talking about
a revis effect
and let's say you were getting a rebus
effect
with sensation so in theory if your
perception is
entailed by the predictions
then it could be the case that
things might seem less vivid to you
it may not be the case that your
perception is more vivid
i don't know if i would say this is
strictly um entailed by the rebus model
but that's one thing
you might think potentially if these
slow rhythms if some of these are
actually
entailing your conscious experience if
these are reduced in their coherence and
their power in theory you might get a
diminution of perceptual vividness
however in a cebus interpretation things
might uh appear more vivid not
necessarily more vertical but more vivid
and so
it seems like a rebus interpretation
would say that things might be more
detailed more accurate um more richly
textured with

more faithful correspondences to the
external world when you're perceiving
but potentially less vivid
not because this is a strict entailment
but that seems like one suggestion
under a cebus interpretation
this would be
more
vivid but not necessarily more precise
it actually might be a little bit
smoothed over like like an auto tuning
or a filtering but um also though uh
more compelling potentially and so this
in theory though would be the
phenomenology of micro dosing in theory
and so
when i get into and so i try to
illustrate this with the brain you know
bigger or smaller synchronous complexes
here um so albus the idea is well we
could have an admixture it's not just
one set of beliefs but we have a
hierarchy of beliefs and in fact a
heterarchy so maybe the alpha starts to
get pulled back and that loses coherence
and that seems to be one of the
strongest associations in the
psychedelic meditation literatures is
alphas what people will emphasize is
some of these uh ego decentering or
dissolving effects
but maybe beta stays the same or even
increases in power and so under a rebus
model for instance uh
the idea would be that you know things
like alpha
go down in power but things like those
particular smaller beliefs they can come

together in different ways and different
sort of creative combinations and so you
might get something like synesthesia
with psychedelics as like a creative
recombination of things um under this
more anarchic state once you took the
the alpha self ruler and you reduced its
its tyrannical grip over your perception
and so this more free-flowing perception
you get all sorts of different creative
combinations uh and consciousness
and so yeah so that's one thing is
trying to basically map different
combinations
of
uh
rebus or sebis effects that you might
expect
under different doses with respect to
either sensation or imagination so
really digging into the phenomenology
and using psychedelics as an opportunity
actually in a way to
indirectly test models of consciousness
but also if we're going to explain what
these substances powerful substances are
doing we really actually need to take
some of these questions head on maybe
even the hard problem
so
some of the more
so in this pre-print um
i also
uh
are getting into okay well what do we
mean by beliefs what kind of beliefs are
we talking about
and so uh so one kind of beliefs would

be so you know here you might have the
beliefs
with active inference and the free
energy principles beliefs the whole way
through oh which belief swimming and so
um you know and rebus is saying what's
specifically being is being relaxed is
your deeper beliefs your
internal working models the higher level
beliefs corresponding to things like
selfhood and its relation to the world
um things like autobiographical and
narrative selfhood these are the things
that are losing their grip
and
and within rebus you know i don't wanna
like there is room for sebas effects and
and some of the ways it's discussed
where it's not just across the board all
the slow rhythms are going down the idea
might be that you know
i've heard it discussed differently and
so
you might get
you know an indirect strengthening as
you remove
these slower integrative and potentially
suppressive processes
but in terms of types of beliefs
if we're going to be talking about
cognition or
if we're talking about something as like
highfalutin as the self
now we might want to say like well what
kind of self are we talking about like
we're talking about like a minimal
embodied self are we talking about an
extended self and even those like

there's all sorts of bad questions like
what goes into the minimal embodied self
and
how it's relating to the extended self
and how many maybe there's different
kinds of those
and so if we're talking about the
construction of self in its relation to
the world um and objectified selfhood
the
the ego the the self as construed over
time and across circumstances we're
going to need a fairly rich account of
cognition and so here i'm
basically
providing an account of sophisticated
affect of inference
or
imaginative planning and policy
selection as orchestrated by the
hippocampal and entorhinal system and so
this is uh drawing on some of the work
um doing with tim verbellen and others
i'm trying to decipher that system but
basically uh you know
you seem to have
uh so here
you know hippocampus can be thought of
in a way as at the top of the cortical
heterokey
when the prediction errors
are not so you as the frictioners are
percolating upwards
being contradicted or stopped by
predictions you stop them as low as you
can as close to the modalities as you
can that's the maximum savings and
that's what you would expect with the

maximally efficient prediction and
learning
um or
but
if you can't predict a pattern if
nothing the genuinely novel that makes
its way all the way up the hierarchy
into the hippocampus as this prediction
error storage place of last resort
and you this
then can be stitched together
to create memory
and specifically what hippocampal and
enter rhino system seems to do is it
encodes this sequence memory
it has multiple functions but one of the
ways in which one of its core functions
seems to be the encoding of trajectories
of the overall organismic system through
time and space
and so if you look at the work of people
like reddish you'll actually see for
instance as this rodent is like in a
maze um
figuring out where it's going to go
you'll see these sweeps of activity
within the hippocampus kind of play out
in different directions across these
place fields or this mapping
of space and then the organism will take
like one branch or another of these uh
potential sweeps of predictive activity
and so uh i probably don't want to get
this here but i have an account of like
the anterior versus the posterior
hippocampus and be qualitatively
different where the anterior hippocampus
seems to be the one that's more

predictive giving you these sweeps
giving you things like counterfactual
processing the anterior seems to be more
closely coupling with the the posterior
censorium and will be more important for
more specific episodic encodings and
maybe more faithful episodic
rememberings
and so
the idea is you have these little
hexagons which corresponding to the
mapping of space these hippocampal place
fields where is the organism within some
sort of space
and then
they're playing out in some trajectory
and so now along this trajectory it is
you're having the prediction errors from
the cortical generator model which can
basically
by
pointing back to cortex can unpacked
unpack at every point in space
um the experience you would expect for
that particular temporospatial
trajectory and this would be an account
of episodic memory and episodic
imagination or the stream of
consciousness if you will okay so here
we are
and i think yeah this is the last thing
i'll
so here we are
with these trajectories being played out
and different experiences being entailed
at each point
as it plays out and they have a given
length it seems that there's um a

extent of this um i think james calls
the specious present or the idea is
there's like a thickness of the moment
we find ourselves in or
this might be a somewhat different idea
but there's
there seems to be three seconds
is seems to be roughly the upper bound
of what we would call a given
extended moment of experience um edelman
i think called like the remembered
present where there's a little bit of
like a prediction and a post-diction and
you're moving back and forth in this
sliding window
frame of experience
and so one interpretation and so within
here as you're doing these different
episodic
rememberings and imaginings that
one thing that the 580 2a system might
do
is
both increase the vividness of what
you're imagining or perceiving as you go
but also allowing you to have
more extended uh rollouts before you
have to reset
where it seems like a roughly three
seconds might be the upper bound
and so
basically i'm i have uh within the
preprint this uh this comic description
this neurophenomological comic where as
you're increasing the dose imagination
is getting closer to perception
and you're getting a longer runway to
work with for this extended present

moment with within which you are
planning and learning and making sense
of things and so it gets a little bit
bigger with the micro dose a little bit
bigger with this threshold dose you know
things are even more vivid you know when
you started out you're just like
kind of moving through the world i'm
kind of awake and kind of conscious i
guess
and then it's like i am like
things are
crisper and a little bit more creative
like
not that much here
but a little crisper
um this would be something like i don't
know like uh
someone does like a programmer in
silicon valley like micro dosing is very
popular it doesn't seem like they're
trying to like trip but they're they're
almost using it like like like a little
bit of a
a stimulant trying to like increase
absorption
whatever they're doing
and so then
so as we move through you increase the
dose some more and now it's even more
vivid and more creative more different
imaginings you're having access to more
more is manifesting in mind but then
after a certain dose
it seemed
what i'm illustrating is that basically
the the coherence starts to go
and you start to get smaller rollouts

they're even more of it than they're
more creative but they're becoming uh
less uh
clear in terms of the sequencing they
make less sense with respect to
achieving particular goals and so i
think something that i should explain
and we might have to talk it out but the
the underlying this is it's okay so
uh these hippocampal resetting events
are occasioned by different things
one is if i go from one room to a next
um if some sort of like really
surprising thing happens that's gonna
trigger this um sharp wave ripple
activity and a re-tiling of space within
which you're doing your sophisticated
inference um this can so something
either
uh so there's there's some evidence that
reward prediction errors if something
really really good happens that might
cause you to basically ramp up the
hippocampal activity encode that you
know generate like the phasic bur help
generate a phasic burst of dopamine
learn it and then uh keep going or if
something really bad happens this might
be an opportunity to pivot what you're
doing is not working let's retile let's
try again
um
and so
this will be a function of both things
like what's the neuram modulatory tone
what are those like the circuit level
conditions like
maybe directly what's your level of 52a

simulation that might influence uh how
long you get before you have to reset
but it could also be explained at the
level of experience itself at the level
of the expected free energy changing as
this emerging cause so as i'm getting
creative
if i get so creative that i'm no longer
making sense of things my expected free
energy if that goes up like that spikes
up because what i just said is really
surprising it's it's it's novel it's
experience vividly and it's not making a
lot of sense
that might be something that would
occasion this resetting
and so the idea is a model of basically
the stream of consciousness or episodic
memory and
episodic remembering and imagining
which becomes
more
uh coherent
more rich more absorbed within this
first regime and then a second regime
that's more properly psychedelic where
it's
very un it becomes
extremely creative so much so that it's
um becomes qualitatively different than
normal experience and even so much so
that like for instance at a certain
point you're not going to be using this
to enhance your job as a programmer you
will be doing this on vacation with
trusted others or something like this
hopefully and this is not something to
take to work that you know people are

not mega dosing at work hopefully or
they will not be working there for very
long
at most workplaces
um
and so
uh
yeah that's basically
and so then coming looping back around
to the beginning the ideas
with a deeper understanding of these
systems that includes some rebus aspects
and some sebis type aspects
we might be able to use these as
inspiration for novel active inference
and machine learning parameters
so
[Music]
potentially you can think of like
getting stuck at a little like people
you know if they're living like a life
of quiet desperation or something
they've gotten stuck at some kind of
local maxima and they're looking to
change things they're looking to get to
kick out of it they're they're not
finding an elegant path to minimize
their expected free energy and so if
you've
found yourself trapped at this
local free energy minima you can use
psychedelics to turbo charge your
sophisticated affect of inference enough
to maybe find a novel solution see
something that you would otherwise
and then i guess the final thing i would
like to point to is
in terms of using psychedelics as a

probe on mind
you might have different interpretations
of what they reveal if you're thinking
of either a strictly sebas story a
strictly rebus story or some kind of
combination of the both of them and so
i'd suggest that for instance you can
think of a good amount of psychedelic
phenomenology
as corresponding to
uh the unmasking of your core priors
and so things like let's say fractal um
fractal phenomenology
that maybe would be a core perceptual
prior maybe even partially evolutionary
partially as a
reliably learnable posterior or
empirical prior for making sense of the
world and this would make some sense in
that if you use fractals as you're
as generating your predictions that's a
very efficient basis set because you can
you can do a lot of different
combinations of fractals and this is
especially going to be efficient if
that's what nature itself is doing
you're using different combinations of
fractal processes to model a world which
is largely structured by different
combinations of fractal type generative
processes
further things like
entity perceptions uh associated with uh
different psychedelic experiences um
archetypal type
experiences
those also
could be thought of as the unmasking of

core priors potentially strengthened
core priors
let's you know because you'd expect from
active inference that you know entities
is something the kind of thing you would
expect
as
a
a very powerful empirical prior because
that's how we survive especially agents
such as us who think through other minds
and where our primary niche is largely
each other
and so
whether we're you know so the details of
psychedelics might actually reveal
these foundational
inductive biases or
foundational lessons
in the learning curriculum and the
structure learning whereby we come to be
the kinds of beings we are
and that
is it for now
okay
thank you adam that was super awesome um
i am like particularly overloaded right
now on like
my 1a receptors
which is like i just want to imagine and
step back
um before i ask some questions i know
daniel asked a lot so i'm gonna let him
like ground me a little bit in reality
and have the first jab
i was just writing a bunch of stuff in
the live chat so definitely anyone can
ask because

um
[Music]
you jumped through many areas so if
anybody has a question they can type it
but let's take a quick breath collect
our thoughts and then blue you can ask
the first question
wait i just passed to you to like ground
me all right so so um
i guess i'm already here um
all right since i'm already here i'm
gonna like
unclearly ask a whole lot of questions
at once and also like try not to fangirl
out too much because this line of work
is really super awesome and unifies a
whole lot of
my particular interests um
so just
really quickly
what um do you think the relationship is
if any between
inference and memory and can we quantify
that through
the brain the brain system
dopamine serotonin like
do you have thoughts on that
i'm sure you do or do you have thoughts
that you can summarize in you know 10
minutes or less
so in terms of memory um
there's
a sense in which
uh
it's all like there's a sense in which
within the free energy principle and
active inference
you know all

like was it
all effective causation is inference
and it's all inference
based on
different kinds of memories so your
belief every belief is a memory and
everything that happens is a kind of
action as a kind of inference
but there's another sense though of like
memory of the time i'm the kind i'm
trying to get to like the focus on the
hippocampal
enter rhino system this orchestration of
the overall
memory system and this encoding of the
general genuinely novel in these
episodes and these frames of
sense-making
that
this
would be
um
a more complex
kind of inference
and a sophisticated one specifically
where you are
inferring the most plausible
a conce
rather than
so so there could be a sense for ins for
instance
in which
you might get a fairly
under some circumstances maybe for like
recent things that that that aren't like
uh i don't know what the shelf life
would be but like
uh

so some of the literature
and a couple weeks couple months i don't
know but like that actually the
hippocampal system through just this
chaining of attracting states along
these trajectories
could point back to cortex and play out
something that's actually like a fair
a decently faithful but still very
biased
account or a reconstruction a
remembering
of what occurred
um but for things that are older that's
probably not the case even for things
that are newer
there's this constructive
so it's always constructive and that
it's always a kind of
every moment perception is inference at
every moment there's a creative aspect
of filling in
but actually like situating yourself
within some sort of causal unfolding
that's going to involve a lot of
value
cannalized policy selection that's going
to involve you using constraints like
well what kind of being do i think i am
and what's plausible and that's going to
determine when you're going down this
like bran this branching forking pass
of different possibilities like you'll
be moving between like well what might i
have done under this or that and you'll
be picking
plausible accounts of you
partially as you go and so

some remembering might be a little bit
um
like the naive like playback one if you
play back the trajectory it's got
pointers back to cortex
and then you have this belief hierarchy
and so you give the prediction errors
back to where things were new and then
it just unpacks itself as a
good enough account of what happened at
each point as he went through some
trajectory and it's a decent
semi-faithful playoff but
there's also going to be this other kind
of memory though
that's that's not faithful at all and
can really deviate
a substantial amount of well we're we're
in hard to distinguish from imagination
and continuous with it it's like how do
you know if you're remembering something
or imagine you use these different cues
one cue you might use
is perceptual vividness so you would
assume like if i remember something it's
gonna be a little bit crisper than if i
just like totally made it up uh the
other one might be like just the
plausibility like yeah that's like the
kind of thing someone like me would do
and but you can get it confused and that
would be a source monitoring error and i
think that's actually in a way for
psychedelics
um both in terms of this core process of
figuring out
who you are where you've been where
you're likely to go and where you ought

to go
thinking of these pathways as
potentially
what it is to be an agent as revealing
ways that agency and
different self-consciousness and
self-hood can vary
but also in terms of this i think needs
to be brought in in terms of what we
might expect to be impacted and what we
might want to prepare for and heart and
and
harness with things like psychedelics so
for instance there might be a very real
risk for phenomena such as false memory
so if the gap between perception
perception and imagi and imagination
starts to narrow if your imagination
starts to seem more vivid it might
become more difficult to distinguish
between
a remembering and an imagining
or um
yeah so things of that nature so
that
i think the issue of what is
memory
um in its relation to inference
is going to be key for what it is like
we want to alter and what it is we maybe
don't want to alter but might
accidentally alter um when we do
different types of interventions so like
you played so perfectly into my
follow-up question was
like how does creativity play into that
right and like what is um
maybe the function of creativity like i

mean personally my autobiographical
memory is awful but like i have a really
stronghold like good factual memory for
like all kinds of ridiculous things
so
so where does creativity play into that
like
memory perception
inference like paradigm and are like the
people that are more creative or that
take more psychedelics like do they have
a less like accurate memory or are they
more likely to remember remember things
inaccurately or even like what is
inaccuracy sorry like let me just get
super meta because i really
fundamentally believe that perception is
different for all of us but like what is
written in my biochem textbook less so
right so maybe that's why like i put
less priority on my autobiographical
memory because it's i believe it's
functionally inaccurate but like
whatever how many angstroms are between
each subunit of dna like that's way more
you know important
right
i mean that's
one of the hardest questions i can think
of
and sorry
one of the most important too um
so there's this i'm trying to forget the
name of the paper it's this dual
constraints model of creativity
um that does address like head on the
issue of kinds of creativity and
psychedelics um i'll post that uh later

but
the
it does seem like for instance
there are
i wouldn't say
necessary
trade-offs with respect to
uh different processes
as to whether they allow for realizing
different types of value so let's say so
with memory or with creativity you might
distinguish between divergent and
convergent creativity so like conversion
creativity where like you want to hit a
particular
target and generate a novel solution for
a particular
challenge
or divergent where let's say you just
want to think of something
new um maybe
just a
novelty generator
from which you then might select later
convergent solutions in a longer process
or just uh as part of an open-ended
exploration
and so
you know there's different ways like you
for instance like
you could
get
if you just turn up
uh the heat the temperature
and you make things more
imprecise
and unconstrained
you might be able to get

all sorts of different combinations and
this might be a good like fountain of
diversion creativity
some of the evidence suggests i believe
that uh
even with the micro dose regime so i
think
people who are doing like micro dosing
usually
they're looking for things like enhanced
conversion creativity i don't think the
evidence is great that that's actually
being achieved
but uh divergent creativity might be
more likely to be associated with
psychedelic experience proper of
combinations of things that just you
never would have imagined it's not clear
it's not necessarily even useful
for any particular thing it's not
hitting any particular target
necessarily it might be after the fact
it might reveal something to you that
you never imagined that made all the
difference maybe like it juked your
defense mechanisms in terms of like your
policy selection was leading into garden
paths you're seeing some things and not
others and then it took the blinders off
it revealed something to you that you
might have missed
but i'd say that people definitely vary
partially maybe due to things like
low-level parameter tunings like
different people will vary in terms of
like the natural level of the gain on
these systems the natural tendencies for
occupying these different regimes

and so you'd expect that to vary either
through some combination of things best
explained at low levels like you know
different alleles of neuromodulators
um and some of it would be not well
explained at that level it's more just
that's the way the structure learning
happen that's the way the system
bootstrap itself different kind of agent
and and everything in between
so
one thing that i find to be like uh
i want to go into is
that humanness in general us as a
creative species
that these pathways might have been
important for allowing us to
have
be more creative and being able to
combine things in
a broader range of ways
and that this potentially uh you know
unlocked a more flexible type of
cognition maybe a greater capacity for
analogical reasoning the kinds of things
that might have potentially precipitated
this great leap forward and the birth of
cognitive modernity where we became a
symbolic species and cultural
evolution came off the ground
there there's an uh older idea by
mckenna called the stoned ape theory
that like there was actually like a
period where we were in this symbiosis
with mushrooms and we're eating
psilocybin containing mushrooms and this
actually
helped to get that going

i don't think we could rule that out
but you could also get something very
similar with a you know a couple high
impactful mutations along these pathways
uh
that doesn't involve a mushroom eating
phase and so there still might be a
really interesting account of actually
the secret of our success as a species
as a particularly creative kind of
hominid who's able to bring things
together in more um
novel imaginative ways
that um
could be involved
in being continuous with the story of
psychedelics and how they work
and in the present day so that obviously
you know i'm extremely interested in and
also as a source you're saying
individual difference like you know
you know are there more like to this day
more shamanic personalities are there
you know are there more
and to what degree can this account be
told at the level of genes or is it or
epigenetics or is it not well told there
uh one
yeah so that's those are some thoughts
on that okay so wait now you just like
walked into this trap that i laid for
you so
i've now got to ask this next question
so you talked a lot about um
psychedelics and sensory deprivation and
now you want to talk about like our
symbiotic relationship with fungi
and i'm a huge like terence mckenna fan

so if you've read the archaic revival
which i'm sure you have
so where do sensory deprivation
in conjunction with psychedelic use like
where does that fall
on this like perceptual
perception action
mental action
alpha
beta brain wave serotonin dopamine
spectrum like i just want to know your
thoughts
sorry not sorry but i'm sorry
so
i don't know if this exactly gets at it
but it's possible that um
this would be rather different but like
okay like this is this
we're really going to go into conjecture
land here
um but
so i think what the book is uh it's a
recent one um
about us is so that there's like two
domestication books recently about
humanness
um
[Music]
so one of them is uh the goodness
paradox and us is a self-domesticated
species then there's this other more
recent one but the idea is that this um
serotonin
pathways
if
it depends which ones we're talking
about will be quite different but
if

they are up regulated unless they have a
1a variety you could think of that as
being more conducive to something like a
domesticated phenotype
so
if your
dopamine is higher and
you're
you're having uh
you're expecting you're more confident
in your policy selection
then
that might also cause you to be somewhat
impulsive
and that might make you not necessarily
be able to
discount utility enough hold back think
things through a bit more and you um if
the dopamine is too high
you might get
less uh cooperativity
if it's and so you can think of the
serotonin to the patients that it's the
1a in particular as being um
of a domesticated variety make and
things might appear for instance less
vivid perceptually um and imaginatively
in theory along
if we're just talking about those
pathways
the 2a
you know
for all this
um there's so much we don't know
so so much of this is conjecture but the
2a does seem to have a little bit more
of a dopaminergic character to it
and

the
and actually it seems like that pathway
evolved
roughly around the time of the cambrian
explosion
um around with the advent of jaws in the
context of predator prey arms races and
so that might be consistent with like
a dopaminergic significance so this like
temporarily like increasing um your
confidence your policy selection like
turbo charging your action selection a
bit so you can go out and get the goal
of either
getting your lunch or avoiding becoming
lunch
and so
let's say you're moving into
greater gain on imagination let's say
we're thinking serotonin either of the
one air 2a is doing this this then might
compete with more externally focused
perception sensation and so you're
imagining more but because your mind's
eye is occupied more with your
imaginings you might be attending to the
details of the environment a little bit
less
and so uh a species or an individual
who might have to gain for whatever
reason higher on these pathways um you
might call this a little bit more
shamanic or maybe a little bit further
on like a schizophrenia type spectrum um
might be perceiving the world slightly
less faithfully
but
their imagination is now freed up to be

more freewheeling and more creative
and this is allowing for
different things to come into being so
very speculatively we might think of
like neanderthals as like less
domesticated humans
maybe even a little bit more like autism
spectrum but like better able of
perceiving the world precisely
um and us we might have been a little
bit more shamanic a little bit more uh
although we are the dander cells also
because we all interbred and the very
you know the variance within is probably
bigger it's definitely bigger than the
variance between
um but
uh
us and my
the our ancestors and
us um
we we might be somewhat somewhat less i
don't know i don't know if i'd say less
dopamine i don't know if i'd say that so
that's what i got for now
thanks adam um
great presentation
just wanted to make two quick notes and
then ask some questions from the chat so
the first is there's no free lunch with
strategy so there's niches that are just
unwinnable
where the phenotype cannot win
and then there's other niches where
potentially a broad range of phenotypes
all can proliferate
so whether or not a given strategy is
successful or under what cases a given

kind of neuromodulatory state
is good or bad by what metric
there's just no
single free lunch or preference for
lunch food then the second point was the
complex gene family point was very
interesting
and you mentioned some other
receptors in the serotonin receptor
family
and
there are many receptors including ones
that don't have nice
clean names like number one and number
two and number three so
at least 14. yeah so how it's exactly
14
investigating function in these
massively interacting and often
redundant like if seven does something
but only when eight is gone what's the
actual
mechanism there so here's some questions
from the chat so first question from a
chat
stephen asked
interested in the approximately three
second window of conscious awareness
adam mentioned does that trace back to
william james stream of consciousness
awareness and the temporal scale scales
of experience
i believe so
so it's kind of like a three second echo
is our experience or what does the three
seconds represent
so um so the empirically
established

phenomenon is there seems to be
stitching things together into one
continuous unfolding there seems to be a
three second upper bound
within this continuous unfolding i
believe there is relevance to
james's species present and edelman's
remembered president of this idea of a
kind of
anticipatory prediction and looking
ahead and a postdictive going back to
you're moving back and forth of yourself
in this unfolding
and the speculative part is
um i am suggesting that this might
correspond to
the
uh
a
hippocampal endorinal tiling of space
either a physical space or a conceptual
space
within which you're doing this and that
the
the reason this upper bound would be the
length of time you have before you need
to refresh the attractors and re-tile
speculative and then and the idea would
be that this would correspond to
something like a these sharp wave
ripples that you'll observe
um
we don't know for sure it's a perfect
lead into this second question and then
i have another comment from cambridge
breaths so
what you just said about ripples so they
wrote wonder if investigating cortical

traveling waves via neural mass
biophysical modeling approaches
in particular
the work of wilson cowan can contribute
to this work if yes how many thanks for
elaborations
timmerman um in collaboration with robin
uh card harris has
done some really interesting work along
those lines i believe
it was with dmt
looking at traveling waves um i believe
i believe in the alpha frequency as
evidence for the rebus models
and so the idea would be that if you are
actually relaxing your belief landscape
you might get
a reduction of like traveling waves of
predictions going down towards the
periphery and maybe greater inward going
prediction error waves
and uh
and so
some work
has been done along those lines already
uh
by timmerman and others that's really
interesting
and as i was saying before so although
i'm like introducing additional factors
i do not think rebus is wrong i just
want to add a couple additional details
of
like
some of the
contributions from modulations of
cortical microcircuitry might be
different you might be able to get rebus

effects from a variety of ways not just
from this desynchronization
but before i forget earlier you're
saying about free lunches
absolutely and i think this is part of
like you know for instance like let's
say you know if i would just crank it
all to the max um
if each one is like has its own function
well you know it'll be depending on the
nature in
both generally and locally so
we're talking about dopamine too much
dopamine you might be impulsive too
little you might not strike iron's hot
and you might miss opportunities
too much 1a or too little 1a you might
be irritable and impulsive too much
this is opposite problems 2a
too much um
it could generate over time a kind of
psychotic predisposition that would be a
loss of a free launch
the one more thing that you're
mentioning is okay yeah it's not just
these two receptor classes uh there's
actually you know 14 conservatively and
so
one thing
i'm wondering though
is whether you get kind of like
this is potentially wishful thinking but
potentially not um
you get the lie and share by considering
a couple
and that there might be like an order in
which
they evolved and a ubiquity

where the ubiquity um would be one sign
for like how important it is to figure
these things out so you know the just
the broader the distribution in theory
the more you get from
explaining the ones with the broadest
distribution and so the 1a and 2a have
this extremely extensive cortical and
subcortical distribution um but that's
not necessarily the case like you could
have for instance at pivotal areas of
the brainstem you know you're right on
top of the core value signals
you know one class of receptor there
that could be the difference that makes
all the difference
um
another aspect
of
the hope that you can you know make
progress incrementally
you know with the goal ultimately you
know we're going to want to get every
single last one of these receptor
classes
is that
the other the additional ones might have
a common
organismic cybernetic significance
but
they're different because
of their different locations
and so that their functional account
might still correspond to let's say
overall levels of serotonin overall
levels
of
some neural hormone or blood acidity or

some other homeostatic state or state of
challenge
and so that the active inference
significance might be the same but
you're tweaking differently in different
neural systems so that the overall
coordinate effect is coherent
with respect to um
you're expected for energy with respect
to realizing organismic value or
minimizing prediction error with respect
to the prediction of you doing the
things you need to do as a successful
being
um
and so one more thing though is i'll
just throw off the 2c idea is
potentially
with these gene i don't know how far we
can go with this but with this idea of
like so with a gene duplication event
once you
have two copies of the gene this allows
you to basically take one of the copies
because you've got a spare that's still
doing the same thing and now this other
one can mutate and can separately be
optimized by evolution to take on a
different function and so the idea is
that there might have been a common
ancestor that looked a little bit more
1a-like but that at the extreme levels
of it during this gene duplication event
this became the 2a receptor and maybe
this goes one step further to the 2c i'm
not sure but maybe we go a couple steps
further with this kind of
gene duplication story and what we're

corresponding to is peeling out and
separately
module in a modular fashion optimizing
for particular regimes of organismic
significance
um
are there gonna be 14 to 17 i'm
skeptical i don't think is that i
suspect it's like maybe three
or four top stories of this kind
and that the other receptors it's these
core
two to four stories
probably three um
maybe
being
achieved by having different functions
locally but in their coordinated
activity
the same story globally
nice there could be a lower dimensional
manifold that
gives a lot of resilience and also
evolvability for the system and learning
and one thing we've talked about before
but it's interesting to think about
is in your presentation you did a really
awesome job of connecting the general
kind of dynamical systems and inference
questions to specific human regions
and then it's interesting to ask okay
well
insects have a different brain layout
they don't have the entorhinal so we
can juxtapose cognitive systems
and learn a lot about which attributes
of the computation are necessary and
sufficient so i wanted to ask which

cognitive systems you thought were
interesting to study in that sense
i i have an intuition and a hope
that may or may not pan out that the
extent of convergence um is fairly
substantial
now things can i think flip in all sorts
of ways you know i wouldn't just assume
this is correct
but let's say you know we're talking
about within an insect um
you know we would potentially look to
the central complex
and you know which which has a lot of
hippocampal and to rhino functions and
so and it might actually be a an even
better system to study in terms of you
would have you know complete access
theoretically
to the whole
you know you'd have a much richer access
to via something like calcium imaging um
and so
it
that could be
looking at something like
the attractor dynamics
of like you know so there's there's
experiments like let's say like take
fruit flies they put them on top of like
a little styrofoam bowl and hook them up
to virtual reality and they have them
like doing different things you look at
the central complex while it's doing
that under different states
of neuromodulation and we might see
common organism significances and a
common computational account

in terms of
the map mapping space and policy
selection now how far back does this go
some of this logic might even play out
at the level of individual cells
and like either like for instance like
1a versus 2a
1a
so low levels of serotonin seem to have
an effect of strengthening cytoskeleton
dynamics and so that would promote like
an organism kind of staying in place
but
high levels actually destabilize the
cytoskeleton and so they'll actually
like literally create plasticity and
fluidity of the morphology and allow the
the organism to move around more and so
some of these accounts
might go all the way back to the like
near the origins of life and individual
cells and metabolism
uh how far we can go i don't know and so
some of these might have even been
repurposed for the
different
social ill insects communicating
although you could see these things
flipping because you know what goes in
the individual doesn't have to be the
same in fact it can because you have
things and one significance on the
individual level that might mean that
has the opposite significance at the
group interactive level so um but
i think it'd be extremely rich to like i
my kingdom for that science to be done
blue do you want to ask a question

i will jump in here actually um
you know when we talk about the origin
of life you mentioned
earlier in your talk that
the
third person perspective
is related to
imagination and creativity and this goes
back to kind of what i was asking
earlier
um
but
is it so much that the third person
perspective is related to
this kind of like coming outside of ego
and looking at yourself you know
from this kind of meta perspective or
is it just fundamentally the dissolution
of the self like how much does an ant
think about themselves or like a
bacterial that's part of a micro like or
a biofilm like how much of of that is
imagination like looking at themselves
as part of a whole or that's just like
myself doesn't matter
i'm part of this bigger unit um
so like if you're
part of a well-coordinated collective
there's a sense with that within active
inference
you will have converged
on
a shared generative model for your
coherent inaction that you're taking
part in and so your role
of what's being enacted
there's a sense in which um
it's not exactly like on some level

in some way there is information of some
kind which is not necessarily egocentric
in fact it's like your role aloe
centrically within the thing
that like even if
on some level it's like what what
you are
you as generative model
if you were to give the generative model
generative modeling process as a whole
of perspective
it's not egoic it's not from necessarily
from a first person point of view even
though the it it's it's from the the
coordination manifold of the of
of the group that's doing this this
common thing
am i kind of getting it yeah
um
yeah that's and
it's interesting like developmentally if
we were to kind of take when these
different perspectives of different
kinds
come online
like for instance right out the gate
it's not clear to me that we have like
for instance coherent egos like it's not
clear we attain to this like right out
the gate when we're born
and so there's like a sense like by
default the
um
you know the infant might be egoless
right it's it's and by default it's
doing this more like inactive coupling
with its caregivers and
this sort of thing and as not yet

actually being um
enslaved to an egocentric point of view
um
there's kind of an interesting thing
with language where it's actually like
the ego itself seems like when we use it
like in like psychodynamic terms is
actually so we say ego central
perspective it's like the i and it's the
first person but when you say like ego
as like a kind of like
extended unfolding of you that's in the
world and all like your possessions and
your sense of yourself within a social
context that actually is more third
persony so it's like ego and from a such
perspective is more first person even
ego
as like psychodynamic concept and the
kind of thing you might want to modify
with a psychedelic
is actually
more of a third seems like a little bit
more like a third-person point kind of
thing but
um and so now the question would be
though like maybe there's like an
implicit
um aloe centricity or non-egoic-ness
um
being enacted by any coupling and maybe
that's like the primal like the primal
the primordial state
um and then you attain to an ego central
perspective but there seems like there's
like another kind
of um
more psychodynamic ego or

objectification of self the third person
um of taking
an explicit
perspective they think
it's it's actually first person and that
you're perceiving it from a point of
view but it's fictitious and you're
taking the view on yourself
as if from without
and that might come later in development
at some point maybe through like
enactments of um mirroring between
uh infant and caregivers or between con
specifics but like if you're mirroring
with others then you might be able to
establish these cross mappings between
what you do and what someone else's does
and this could potentially help you
to create representations that you can
then harness for viewing you as if from
the outside
but
when we're talking about so when we say
self when we say ego like what kind of
ego what kind of self
but
it seems like
there's a
primordial like minimal self
egocentricity of just seeing from from
behind your eyes but then there's more
like a psychodynamic ego uh slash me
uh which is involving the
objectification of you adopting a
perspective on you
imaginatively
as if you were viewing yourself from
without and then contextualizing

yourself across space and time and in
different contexts different social
unfoldings as
what you would need for something like
an autobiographical self
something of that nature
well so
how does an ant see it though does an
ant have like is the first person gone
or like does is an ant entirely meta all
the time or a bacterium is that like a
total meta perspective and they don't
have to go through this like ridiculous
like self like
whatever creating a dissolving loop or
danny would know better than i would
you actually have to ask the ants that's
kind of the hard part but we can make
one comment the difference between the
egocentric and then the more impartial
or autobiographical reminded me a lot of
the projective consciousness model
projective geometry model which was also
integrated with higher order theories
and global workspace and a lot of those
other
topics with
um ryan smith and christopher white
and also the more geometrical work um
with rudraf and others because it's like
we do need both those modes we need the
one where the hands are bigger when
they're closer to your face and we need
the one where it's like
you're playing the video game with the
third person view
and then it was really interesting when
you point to a lot of the um

seen as deficits of memory or cognition
like false memory but just seeing them
as computational outcomes of
stochastic interoceptive
lossy compression and generalization
semantic encoding
what's more likely who kind of person
are you that's gonna
you know scale whether you think you did
something or not
so that's just such a interesting area
that you kind of took it down
thank you
it's interesting bringing up like
rudolph cause like there's some things
like i wonder like you know when do you
first get
the capacity for projected geometric
modeling of what kinds
and so like
this so with and what's the nature of
the projective geometric model like is
this something that should i still i can
stop the sharing
there we go um is this something that
you would expect
uh
at the level of like
cortical micro circuitry like some
interpretation of like what cortical
macro columns are doing or is this
something that's actually more complex
and it's like a distributed algorithm
like hinton has like ideas like glom
that kind of get it something like this
like the because what you have with this
projective geometric model is the
ability to flexibly move

between egocentric and allocentric
perspectives i can take your point of
view or mine once i um raise everything
or boost it up into this um
i forget that is it hyperbolic or if
it's whatever this the space is that he
puts it in
from within there you can embed either
either one and move between them um the
thing i'm wondering is like the nature
of these like you know i just mentioned
this uh like
ability to
um
take fictitious points of view on you
i'm wondering whether like the way in
which we do it is a lot more
derived and a kind of skill
of moving between
um
things like the representations you
might have acquired through mirroring
with others
and it's and like to to take another
perspective rather than it being
something that that the visual
system can just do of moving between of
automatically translating from points of
view within this higher space like what
is the nature of that model like i've
even wondered like for instance is
perception actually
always 2d
and in a kind of like holographic way
it's always a 2d projection and when we
talk about something like depth
it's actually either some sort of like
uh

sense of affordance or
intercept of cross mapping or the likely
transitions between 2d or even like 3d
like we like you know we never like mars
said it was like 2.5 division like is
that just the ability to make sure that
what's on the screen
that's 2d is always
coherent or that you move between as if
you can move around a thing but we
actually never perceive anything in 3d
um in terms of our visual spatial
awareness
but to loop around to the ant
um i don't know but like you might
distinguish for instance
if you consider the perspective of the
ant itself as generative model
then
that might be
by default
more allocentric especially because of
its like what it's specifically doing
to the you know it takes something even
makes sense to pick it it's
it's fundamentally part of the super
organism the organism is it's not even
super organism like like what like
daniel says like it's it is the unit
that makes sense it's being selected is
the shared enactment
but then if you go maybe like the
attractor dynamics of like the central
complex it's like quasi hippocampal
quasi-thalamic thing
that might be more egocentric
and i don't think you're going to get
just totally agree with that i think

it's an established fact that
um the compass heading or the
polarization of light is processed
within one nest mate head and then there
are collective decisions like the sex
ratio of the larva that they're going to
keep or the overall regulation of
foraging that
wouldn't be expected to have
even
a correlate necessarily in the
head capsule of any single nest mate
and so i think that's where the
discussion comes to like multi-scale
systems which
is kind of a fun topic because in
multi-scale systems we see sometimes a
clean mechanistic difference between the
lateral and the kind of vertical or the
nesting of systems so like communication
we can see the interface where you have
the lateral layer of
individuals and then where you have the
layers that are
inside of the individual
it's a little harder to disentangle it
if they're all like kind of
wire but
um
what are you or blue do you want to ask
a question otherwise yeah i mean it just
makes me think of how like are we
fundamentally broken with our like
self little egocentric loop right like
that
are we maybe like measuring forever the
incorrect scale of
stuff like should we

maybe be more concerned about like the
scale of all of us together
or the scale of all of the parts that
make up us instead we like kind of
falsify and create like our ego and our
relationship we like project a lot about
how we think we should be within the
world or how we think we things are
within us that like we don't actually
know anything about
i don't know just related to that
i mean
it's definitely beyond neuroscience but
like i'd say uh
yes
but also um
like i both think this is one of the
most valuable and fraught parts of uh
psychedelic and
discourse around psychedelics and
meditation also
where it's like if you get this myopic
egocentric perspective that's gonna be a
real problem
um for multiple reasons it's even for
you it's gonna be a problem because you
see things from one way you're gonna get
owned by yourself as a kind of like
you'll get trapped in all sorts of local
you see things in one way you really
barely see it anyway um
but
i do worry a little bit though about um
overly
you know egos get themselves in trouble
but like the ability to stitch together
for instance
you know there is like a unit over which

you have potentially maximum purchase
and then and like in terms of influence
there's a sense in which like you're the
steward of that because like the causal
closure is densest
and there's also a sense in which like
let's say um
i want to coordinate with others
um
including myself across time like the
ability to like have a narrative
structure
of the ability that's going to be
helpful for not like just jumping at
whatever
source of utility or or expected free
energy comes along and so like to
basically avoid being um
functionally psychopathic like you kind
of need um the ego but you also need to
have the ego not be too tight to avoid
being functionally psychopathic
so it's like not too tight not too loose
otherwise
the whole thing's wrecked
that sounds like narrative stigmergy
like agent environment feedback loops
and
also of the script
research
that we've talked about previously
with alder russen and others where
that's sort of the top down prior so
it's related to what you said about the
hard problem as to if or why or why not
there would be
awareness of experience
at for example the bodily level versus

the you know tripartite zoom chat level
um but at the very least instrumentally
those top level scripts
are the narratives that decide like are
we going to put everybody you know
through a a tunnel where only this exact
width is going to make it through or
is that a phenotype that we have a lot
of
ambiguity around
using the bayesian prior term so um
it is interesting how
and i almost want to ask you whether you
started with the more neuroside
and found it broaching to these
questions or whether you started maybe
focus on other areas and then saw the
brain's specifics as increasingly
important
like there's so little separation for me
between like
the neuroscience obsession and the
existential like i'm constantly doing
i'm constantly in a state of existential
freakout and neuroscience obsession that
i don't know which led which i don't
know who's the dog who's the tail and
who's wagging whom it's just constant
existential crisis and
obsession
wait so i don't believe
what would be
advice to somebody who's curious about
the field or they want to be
working in an interesting area
of neuroscience and embodiment and
everything
that you kind of mentioned today

it seems like if someone wants to focus
on psychedelics
the best
route is to uh
cut your teeth or to like basically
get a home outside of psychedelics
establish a competency there and then
try to bring that within the psychedelic
domain that seems like the best path
that um
most people who succeed in the space
take
um
and
yeah so it's and you know the standard
fraud thing of academia where it's like
you're trying to
not be an inch wide and a mile deep and
you're trying to
not be a jack of all trades master of
none
and where
the perverse incentives abound and
uh
good luck
um uh like
some combination of like
both paying your dues and not optimizing
yourself into
a brittle straight jacket where you
can't see anything past the next grant
yikes
i feel you
thanks for the answer so one small
question from the chat and then if
anyone else has any final questions so
adam thanks for your time and the great
presentation and cool work and hanging

out with us so
adam mentioned the dmt paper with
respect to the cortical traveling waves
the question was wonder if there's any
other work on non-alpha waves and then
just
if you know about that and then just a
little more generally why do these
traveling waves matter or what
matters if they are important or not
so
there's a sense in which um
like i drew like those swaths those like
synchronous complexes but those are
necessarily a core screening you zoom in
on them and it's like so what's the
ontological status it has a certain
relational realism and that the extent
that makes sense like draw that out it's
like
active inference in general with markov
blankets it's like you can it's relative
to what other system that's engaging in
what types of inference they'll just
decide like the scope of relevance i
mean there's additional like autopoetic
self-organizing elements of like in
the internal but
uh so if you look though with like
a synchronous like an alpha like beta
alpha you zoom in it's actually
comprised of these traveling waves as
more of a fine-grained description
although you zoom in on that that's
actually corresponding of population
level activity um and so
it'll depend so there's going to be
potentially if you're looking at

traveling in cortical waves
that might be useful
so i focus on like the standing wave
description as like
um a joint belief over whatever is in
its scope but if you're talking about
something like predictive processing
um
within the model you you might want to
zoom in on this standing wave or
the standing of description for like the
business end of passing on your
predictions and then kicking up your
prediction errors but depends like
it depends on the the the
within the multiscale account what's of
interest for the modeler
uh
one quick thing from this in the side uh
handle the alpha wave question whether
there's any non-alpha wave traveling
wave papers
i don't think so i think the one
i think the terminal one was focusing on
alpha if i'm remembering i could be
misremembering i think
but there are four psychedelics um uh i
think
terry
sanowski i think would be the person to
look at for some really good general
biophysics of traveling waves in the
brain
uh
yeah sanowsky i think would be the place
to look a quick aside on on dmt
like i mentioned like there might be
endogenous cmt um there's

debate over how significant it is so
some people say
it's not significant at all some people
say it's very significant so it's one
place we know that it's produced is in
the pineal body the seed of the soul
and it's
uh
another place and there's some
evidence but some people have questioned
it that it might be um almost akin to
like a neurotransmitter system in its
own right we just didn't know how to
look at it before um jimoh
borgen at university of michigan if
you're interested in that issue
and that debate uh but it's possible
though like
um one of the main endogenous ligands we
don't know
of the five hd2a system one of the main
stimulus might not actually be
serotonin but might be dmt so like under
like let's say like a limit experience
like uh
you know so one like i mentioned like
carbon dioxide can do it or maybe lactic
acid like you're you're running for your
life or for your lunch and it's super
high levels of blood acidity and that
might be activating these
proto-psychedelic pathways um but
another one could be like potentially
dmt mediated and things like um
sex
earthing and near-death experiences so
like you almost bled out
um maybe it's time to relax your beliefs

you're
about to bond with another creature for
years maybe this is how part of how you
do it um
whether bonding with a
mate or your offspring
very expected these are early days
interesting and also made me think about
brain and body dopamine and serotonin
other neurotransmitters in the blood
in the blood vessels the immune system
and then neuropeptides the whole
continuum of short
acting neurotransmitters to slower
acting
phenomena and then
oh we're thinking about it from a nested
multi-scale systems perspective with
this kind of more local faster
and then slower
broader spatial scales which just has
a lot of analogy with everything from
regional governance to
the regulation of tissue so it's kind of
an interesting idea how
a lot comes together there
it also made me think of how
staring into someone else's eyes for
like a prolonged period of time induces
like violent disturbing hallucinations
which is like reported in
i don't know some paper i can't remember
but i'll i'll look it up and and see if
i can post it in the chat but i'm really
interested
how like thinking through others minds
or like looking through others eyes
um is is so violently incorrect to us

that it's perceived as like an altered
perception
i i wonder if this is speaking to like
the way in which we
bootstrap um
like objectified selfhood and and me and
like but but because like we do it so
much this is like a core axis of
imagination and perception
that you can create like these
weird feedback loops and i also wonder
like how much it varies across people
like
um
like like if you're like really far on
the autism spectrum like
or or far in a schizophrenia spectrum
like would you like in a moment like you
look in someone's eyes and it's like it
takes you two seconds to enter like the
psychedelic space
or might take like you know you got to
sit there for like an hour okay
this test was like i think 10 minutes so
i'll i'll find it right now post it in
the chat it's happening to me right now
well adam not your first
active livestream not your last so
thanks for
joining and sharing this
cool work
we hope to
see you around the lab and just in the
area so
always a pleasure and an honor talk to
you later